Membrane Air Separation

Pre-Lab Plan
Unit Operations Lab I
November 14th, 2019

Group 1, Thursday, PM section
Alexis Miranda
Dulce Colindres
Kathya Chavez

1. Experimental Objective

This experiment revolves around the separation of air into oxygen and nitrogen components using a membrane separation apparatus. The goal of this lab is to evaluate the overall transfer coefficients K_{O2} and K_{N2}, separation coefficient αAB, the experimental separation factors, $\alpha'AB$, $\alpha'AB$, and the O_2 and N_2 recoveries, all as functions of operating pressures measured in lab.

2. Experimental

The figure below shows the apparatus used for the membrane air separator. Compressed air from a tank, enters the system with various valves and meters for testing the quality of oxygen compared to air.

2.1 Apparatus

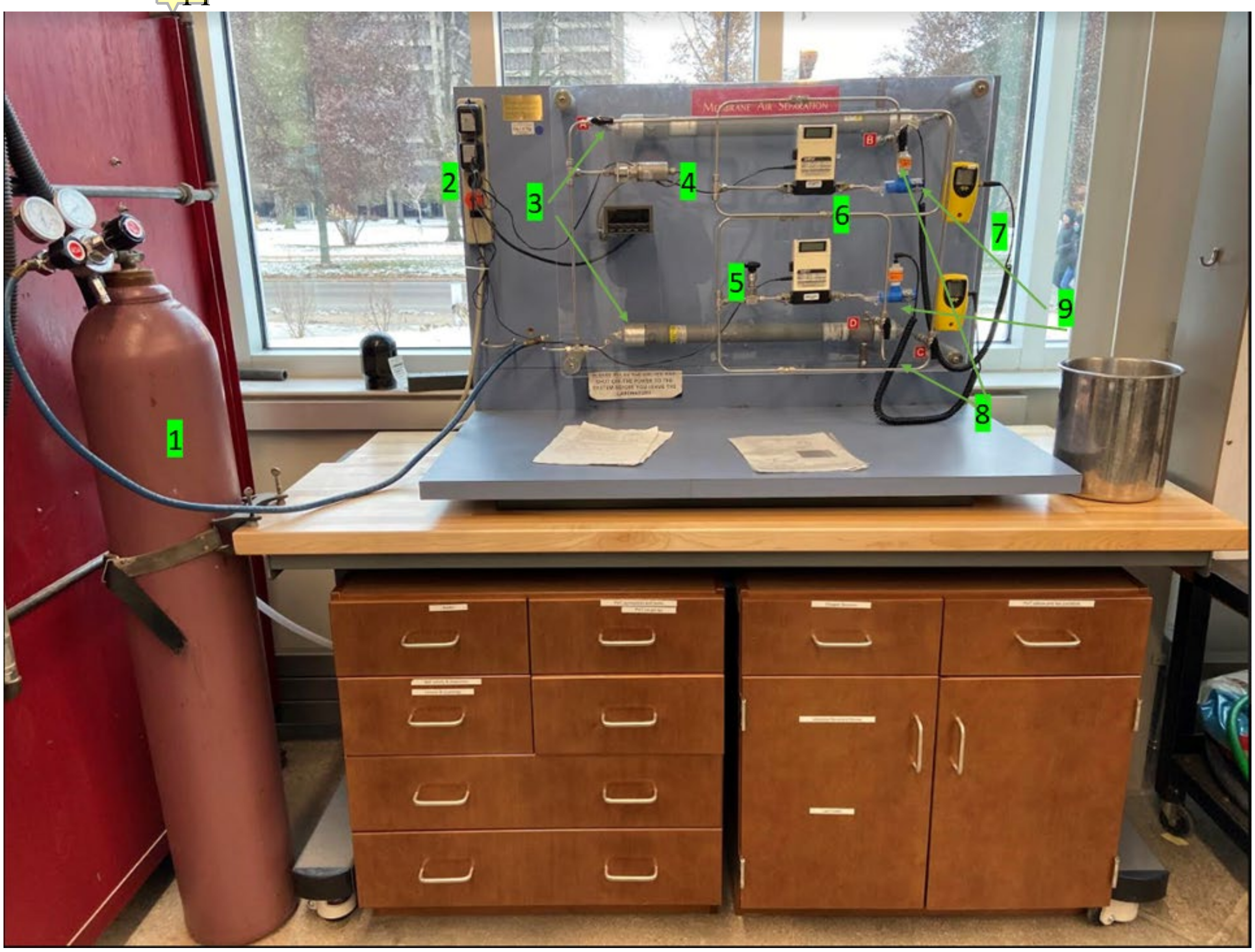

Figure 1: Apparatus for Membrane Air Separation

Table 1: Apparatus Components

Component	Description
1. Air Cylinder	Pressurized air for testing
2. Power source	System Power source
3. Two PrismTM separator modules	Columns with semipermeable membranes
4. Pressure Sensor	Reads pressure flow
5. Needle Valve	Control Flow
6. 2 flow meters	Sensor to read quality
7. 2 O2 sensor	Reads oxygen quality
8. 2 Exist tubes	Allows air to flow out of the system
9. 2 filters	Allows Air sensors to read air quality

2.2 Materials and Supplies

Table 2: Materials and Supplies

Material / Supplies	Description
Batteries	To power the oxygen analyzers
Compressed Air	Compressed Air provided from tank next to apparatus
Oxygen Analyzers	Provide recovery rate, shows when system at steady state, battery powered
Mass Flow Meters	To provide flow rate data, powered through power strip attached to apparatus
Pressure Gauge	Provides pressure reading in system, attached to apparatus

2.3 Experimental Plan

1. Set the outlet pressure to 140psig by opening the valve on the high pressure air cylinder and adjusting the cylinder regulator.
2. Turn on oxygen analyzers and allow time for the readings to reach steady state. Calibrate if necessary.
3. Make sure the mass flow meters are plugged into the power strip along with the pressure transducer display.
4. Parallel Configuration:
    a. Point valves into the correct directions:
        i. VA- Left, VB-up to sample and down to vent, VC-down, VD- up to sample and down to vent.
    b. 5 runs will be conducted at 20psig and varying flowrates, 4 runs will be conducted at 40psig with varying flowrates, and 4 runs will be conducted at 80psig with varying flowrates.
    c. The runs end when the flowrates and oxygen levels are constant
    d. Changing flowrate may change pressure so readjust pressure as needed.

e. Record operating pressure, tube side flowrate, tube side oxygen level, shell side flowrate, and shell side oxygen level

5. Series Configuration:
   a. Point valves in correct directions:
      i. VA-right, VB-up to sample and down to vent, VC- up, VD- up to sample and down to vent.
      ii. Run and collect data for 15 runs, similar to parallel configuration

Independent & Dependent Variables

Dependent
Oxygen Levels
Independent
Pressure
Flowrate

Data Collection

Operating Tube Side				Module 1	Module2	Module 1	Module 2
Run #	Pressure (psig)	Flowrate (mL/min)	Tube side oxygen level (%)	Shell side flowrate (mL/min)	Shell side flowrate (mL/min)	Shell side oxygen level %	Shell side flowrate (mL/min)
1	20	500					
2	20	1000					
3	20	2000					
4	20	4000					
5	20	8000					
6	40	1000					
7	40	2000					
8	40	4000					
9	40	8000					
10	80	1000					
11	80	2000					
12	80	4000					
13	80	8000					

This table will be filled out during exSperiment, thus presented in final report.

Project Safety Evaluation

Project Title: Membrane Air Separation *Date: 11/14/2019*

	Yes/No	
Potential electrical hazards?	Yes	Power strip next to apparatus powering machinery which can cause electrocution.
Potential mechanical hazards?	Yes	Pipes with varying pressure. Pressure buildup in pipes can cause them to burst
Potential pressure hazards?	Yes	Compressed Air tank is used. All safety precautions when using compressed fluid in tanks should be taken
Potential temperature hazards?	No	Experiment will be conducted at standard room temperature.
Hazardous/toxic chemicals involved?	No	Compressed air in closed system would be used but not any toxic chemicals.
Gloves required?	No	Always recommended but not required for this lab
MSDS reviewed by all team members?	Yes	MSDS reviewed by all team members
Process Parameters	**Max Expected**	**List location and possible hazards and precautions**
Temperature (ºC)	25ºC	Room Temperature
Pressure (psia)	140 psig	Lab will be conducted at varying pressures, with 140 psig chosen as our max
Safety Controls	**yes/no**	**If yes, Provide description**
Wearing proper PPE, being careful when inserting particles, wear glasses, gloves, face masks if possible	Yes	Wear goggles

I have read relevant background material for the Unit Operations Laboratory entitled: "Membrane Air Separation" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Alexis Miranda — *11/14/19*
Signature of team member — *Date*

Dulce Colindres — *11/14/19*
Signature of team member — *Date*

Kathya Chavez — *11/14/19*
Signature of team member — *Date*